

IMPROVING DEXTEROUS PIANO PLAYING POLICIES IN PIANOmime

DRL Course Project Group 28

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ABSTRACT

We study dexterous robotic piano playing in PianoMime, where two anthropomorphic Shadow Hands execute target key-press trajectories on an 88-key piano. We reproduce the original single-song specialist policies and the multi-song generalist policy, then evaluate targeted modifications for note-level F1. In the specialist setting, onset-aware reward shaping, light smoothness regularization, and residual factor calibration improve F1 on `Happy_8` from 0.8725 to 0.9187. In the generalist setting, a motion-weighted low-level diffusion fine-tune improves the five-clip mean F1 from 0.6150 to 0.6258. Smooth denoised diffusion training, goal side-channel conditioning, and reward-based rollout selection also produce positive gains. RPIM-based integration gives useful diagnostic evidence and shows unstable average gains on the selected five-clip benchmark. The results highlight the role of note timing, smooth hand motion, goal conditioning, and representation alignment in dexterous piano playing.

1 INTRODUCTION

Robotic piano playing provides a compact benchmark for high-dimensional dexterous control. The agent controls two anthropomorphic robot hands, observes robot state and future note goals, and produces joint-level actions or residual corrections. The task requires spatial accuracy, contact timing, and multi-finger coordination. A policy can generate plausible finger motion and still lose note-level F1 through missed onsets, weak presses, or wrong-key contacts.

RoboPianist introduced simulated bimanual piano playing as a goal-conditioned reinforcement learning task with anthropomorphic hands and a full-size keyboard (Zakka et al., 2023). PianoMime extends this line by using internet demonstrations and paired MIDI files to train a generalist dexterous piano player (Qian et al., 2024). The PianoMime pipeline extracts fingertip motion from videos, converts MIDI into piano-roll goals, uses inverse kinematics to obtain nominal Shadow Hand trajectories, trains song-specific residual RL experts, and distills the resulting demonstrations into a two-stage generalist policy. The high-level policy predicts future fingertip trajectories from MIDI goals. The low-level policy maps fingertip targets and robot proprioception to hand actions.

Our project follows the Dexterous Piano Track. We first reproduce PianoMime baseline behavior for both single-song specialists and a multi-song generalist policy. We then study modifications motivated by the observed failure patterns in the reproduced setting. The specialist experiments focus on reward shaping, motion smoothness, and residual scaling. The generalist experiments focus on low-level diffusion training, explicit goal conditioning, stochastic rollout selection, and RPIM-based data integration. The report uses a direct organization: Section 2 introduces the task and prior systems, Section 3 presents the complete method design, Section 4 reports results, and Section 6 discusses the main findings.

Our main contributions are as follows. First, we reproduce three PianoMime single-song specialists and a generalist policy on five unseen clips, with F1 curves, quantitative metrics, and videos. Second, we improve a specialist policy on `Happy_8` by combining onset-aware reward, light smoothness regularization, and residual factor calibration. Third, we compare multiple positive generalist modifications under the same five-clip benchmark and identify motion-weighted low-level diffusion fine-tuning as the best average improvement in the merged reader-facing comparison. Fourth, we report negative and diagnostic findings showing that naive smoothing, coarse song-level conditioning,

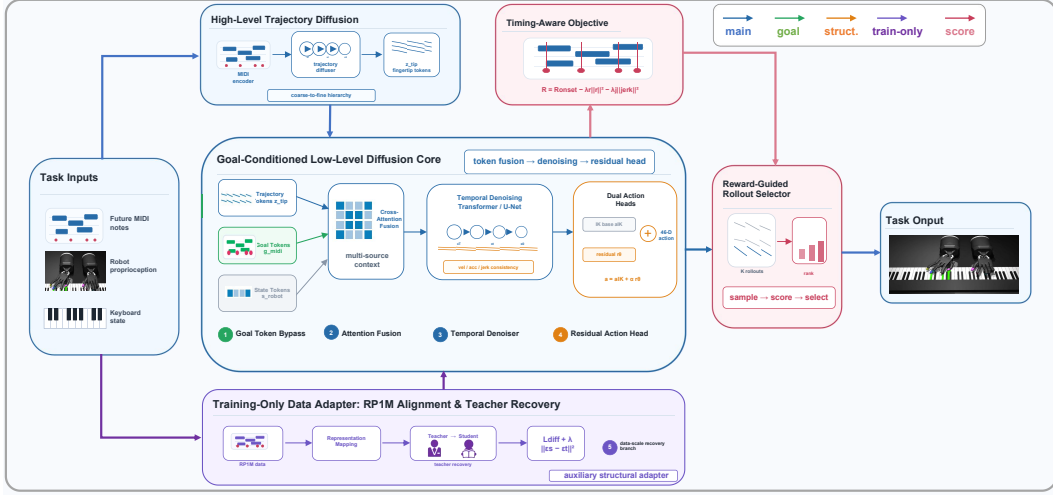


Figure 1: Architecture of our method. Future-note tokens bypass high-level trajectory prediction; multi-source attention fuses trajectory, goal, and robot tokens; a temporal denoising backbone generates smooth action sequences; and a calibrated residual head combines IK actions with learned corrections. Dashed purple paths indicate training-only RP1M alignment and teacher recovery.

and unregularized RP1M recovery can hurt note-level F1, highlighting the timing-sensitive nature of dexterous piano playing.

2 BACKGROUND

2.1 ROBOTIC PIANO PLAYING AS AN RL PROBLEM

Each piano clip is an episodic control task. The environment contains an 88-key spring keyboard and two Shadow Hands. The policy observes proprioception, key states, and future note targets. The action is either direct hand control in the generalist setting or a residual correction in the specialist setting. The reward includes press success, mimic tracking, timing terms, and smoothness terms. The evaluation uses note-level precision, recall, and F1, plus qualitative performance videos.

This task combines several properties that make standard robot control difficult. The action space is high-dimensional because the policy controls two dexterous hands. The state evolves through contacts between fingertips and piano keys. The target trajectory is sparse in time because a note-level mistake often occurs near the press or release moment. A note may require preparation several frames before the visible key press. A small action change can move a fingertip away from the correct key or shift the press time. This property explains the emphasis on both quantitative F1 and video inspection in our experiments.

The single-song and generalist settings stress different parts of the problem. In the single-song setting, the policy can specialize to one fixed target trajectory and use residual control to refine a nominal motion. In the generalist setting, the policy must condition on future note goals and produce actions for unseen clips. This setting requires stronger goal-conditioned behavior because similar hand states can require different future actions under different local note patterns.

2.2 ROBOPIANIST, PIANOMIME, AND RP1M

RoboPianist trains simulated anthropomorphic hands to play piano pieces with deep reinforcement learning and provides an environment, benchmark tasks, evaluation metrics, code, and datasets (Zakka et al., 2023). It frames piano playing as a high-dimensional dexterous control problem with spatial precision, temporal precision, coordination, and planning. This formulation motivates the use of note-level metrics together with motion-quality inspection.

PianoMime builds a generalist agent from internet demonstrations (Qian et al., 2024). It uses paired piano videos and MIDI files. The video stream provides human fingertip motion. The MIDI stream provides task-level key-press goals. The data preparation stage extracts fingertip trajectories from videos and converts MIDI into piano-roll goals. Inverse kinematics then produces nominal Shadow Hand joint trajectories. Song-specific residual RL experts refine these nominal trajectories using rewards for key accuracy and fingertip-style tracking. The expert rollouts provide robot states, goals, fingertip trajectories, and actions for generalist policy distillation.

The PianoMime generalist policy has a hierarchical structure. The high-level policy predicts future fingertip trajectories from MIDI goals. The low-level policy converts predicted fingertip trajectories and proprioception into hand actions. At inference time, the agent uses MIDI goals and robot state as inputs. The original internet video is used during data construction and expert generation. This design links human demonstrations to robot control while keeping the final policy executable from task goals.

RP1M provides a large-scale motion dataset for bimanual robotic piano playing with around one million expert trajectories over around two thousand music pieces (Zhao et al., 2024). It uses optimal-transport-based fingering to automatically assign fingers to target keys, trains song-specific RL specialists, and collects their rollouts as robot state-action demonstrations. This dataset is closely related to our data-integration experiments because it provides robot demonstrations at a larger scale than a small set of PianoMime expert rollouts.

OmniPianist further scales this idea with more than two thousand specialists, RP1M++, and a Flow Matching Transformer for large-scale imitation learning (Chen et al., 2025). Flow Matching provides a generative modeling framework for learning continuous transformations from simple distributions to data distributions (Lipman et al., 2023). In our project, RP1M and OmniPianist motivate the attempt to use external robot piano-playing demonstrations for improving the PianoMime-style generalist policy.

2.3 DIFFUSION POLICIES FOR ACTION GENERATION

The PianoMime generalist policy follows a diffusion-based action generation pipeline. Diffusion models train a network to recover clean data from noisy data through a denoising objective (Ho et al., 2020). In robot learning, Diffusion Policy represents a visuomotor policy as a conditional denoising diffusion process over actions (Chi et al., 2023). This formulation is useful for high-dimensional continuous actions and stochastic action generation.

In the PianoMime low-level policy, the diffusion model predicts a 46-dimensional hand action conditioned on fingertip targets and proprioception. The model samples an action through iterative denoising. The denoised action estimate can be viewed as the clean action predicted by the model at a given diffusion step. This background is directly connected to our smooth denoised diffusion objective, which adds trajectory consistency terms to the denoised action estimate during training.

The stochastic nature of diffusion policies also motivates test-time rollout selection. Multiple samples from the same conditional diffusion policy can lead to different hand trajectories and different note-level outcomes. Reward-based rollout selection uses this stochasticity by sampling several rollouts and selecting the one with the best reward or F1 proxy.

2.4 PIANOmime BASELINE POLICY STRUCTURE

Given a MIDI goal and robot state, the high-level diffusion model predicts future fingertip trajectories. Given the fingertip trajectory and proprioception, the low-level diffusion model predicts a 46-dimensional hand action. The default evaluation uses the official high-level checkpoint, low-level checkpoint, and autoencoder checkpoint. In our reader-facing multi-task benchmark, all methods are evaluated on the same five unseen clips: *Alone_1*, *EyesClosed_1*, *Hope_1*, *SomewhereOnlyWeKnow_1*, and *NoTimeToDie_1*.

Table 1: Overview of insights and corresponding method components.

Observation	Method component	Setting
Note onset frames strongly affect note-level F1	Onset-aware reward	Specialist
Motion stability affects contact quality and execution robustness	Light smoothness regularization; smooth denoised diffusion objective	Specialist / Generalist
Residual strength controls the amount of learned correction	Residual factor calibration	Specialist
Future note patterns disambiguate low-level actions	Goal side-channel representation	Generalist
Stochastic diffusion rollouts have different execution quality	Reward-based rollout selection	Generalist inference

3 METHOD

Our method design is organized around five observations from the reproduced PianoMime setting. First, note onset frames have a strong influence on note-level F1, since a delayed or missed press at the beginning of a note can make an otherwise plausible motion fail the target key-press trajectory. This observation leads to onset-aware reward shaping in the specialist policy. Second, hand motion stability matters for both physical plausibility and generalist execution. A policy can use locally aggressive corrections to match notes, while such corrections may create unstable finger motion and fragile contacts. This observation leads to light smoothness regularization for the specialist policy and a smooth denoised diffusion objective for the generalist low-level policy. Third, the residual strength in the specialist policy controls the balance between the nominal IK action and the learned residual correction. This observation leads to residual factor calibration. Fourth, the same hand state can require different future actions under different upcoming note patterns. This observation leads to goal side-channel representation. Fifth, the diffusion policy is stochastic at inference, and different sampled rollouts can have different note-level quality. This observation leads to reward-based rollout selection.

3.1 SPECIALIST COMBINED METHOD

The specialist branch focuses on Happy_8. The reproduced baseline already generates recognizable piano-playing behavior, while remaining errors appear around note onsets, weak presses, and residual correction scale. We use a combined method with onset-aware reward, light smoothness regularization, and residual factor calibration.

The onset-aware reward increases the weight of frames where target notes begin. Let R_{task} denote the original task reward and let R_{onset} denote the additional reward on note onset frames. The specialist reward is written as

$$R = R_{\text{task}} + \lambda_{\text{onset}} R_{\text{onset}} - \lambda_{\Delta q} \|\Delta q\|_2^2 - \lambda_r \|r\|_2^2, \quad (1)$$

where r denotes the residual correction. The smoothness and residual penalties reduce unnecessary rapid corrections. The coefficients are kept light because dense notes still require fast finger movement.

The residual-control structure follows the PianoMime specialist pipeline. The policy starts from an IK-based nominal action and applies a learned residual correction. We calibrate the residual contribution at execution time:

$$a = a^{\text{IK}} + \alpha r, \quad (2)$$

where a^{IK} is the nominal IK action and α is the residual factor. This calibration controls the correction strength. A small residual factor keeps the policy close to the IK prior. A large residual factor increases the influence of learned corrections. The selected combined method uses the residual calibration reported by our single-song experiments.

3.2 SMOOTH DENOISED DIFFUSION OBJECTIVE

The generalist low-level policy predicts hand action chunks with a diffusion model. Local inconsistency inside an action chunk can produce unstable finger motion and unreliable contact. Direct test-time action smoothing can delay or suppress key contacts, so the smoothness constraint is applied during training on the denoised action estimate.

We keep the original diffusion noise-prediction loss and reconstruct the denoised clean action estimate $\hat{a}_0$ from the predicted noise. Auxiliary trajectory losses are applied on low-noise diffusion timesteps. The objective is

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{noise}} + 10^{-2} \|\Delta \hat{a}_0 - \Delta a_{\text{demo}}\|_2^2 + 5 \cdot 10^{-3} \|\Delta^2 \hat{a}_0 - \Delta^2 a_{\text{demo}}\|_2^2 \\ & + 10^{-3} \|\Delta^2 \hat{a}_0\|_2^2 + 10^{-3} \|\hat{a}_0 - a_{\text{demo}}\|_2^2. \end{aligned} \quad (3)$$

The first auxiliary term matches first-order action changes. The second term matches second-order jerk patterns from demonstrations. The third term penalizes unnecessary absolute jerk. The fourth term anchors the denoised action estimate to the demonstration action. These terms shape the denoised action chunk while keeping the diffusion sampling interface unchanged.

3.3 GOAL SIDE-CHANNEL REPRESENTATION

The low-level generalist policy receives proprioception and future fingertip targets. Similar hand states can require different actions when the upcoming notes change. We therefore add an explicit goal side-channel to the low-level conditioning. The side-channel provides future-note information to the action model and helps the policy use local note structure during action prediction. This component targets the goal-conditioning bottleneck of the generalist policy.

3.4 REWARD-BASED ROLLOUT SELECTION

The diffusion generalist policy can sample multiple rollouts under the same clip and checkpoint. These rollouts can differ in contact timing and note-level accuracy. We use reward-based rollout selection at inference time. The official high-level and low-level checkpoints are fixed. For each clip, multiple stochastic diffusion rollouts are sampled, and the rollout with the best reward or F1 proxy is selected. This component changes the inference selection procedure and keeps the model weights unchanged.

3.5 RP1M RECOVERY WITH TEACHER REGULARIZATION

The data-side branch studies whether RP1M-derived demonstrations can improve the PianoMime-style generalist policy. We evaluate two variants. The first variant uses RP1M-derived data to replace or initialize the low-level policy. The effective preprocessing includes point-major fingertip conversion and a learned linear mapping from RP1M representation to PianoMime-style representation. The second variant starts from the RP1M-derived low-level model and performs recovery on the original PianoMime low-level data with teacher regularization.

The recovery loss is

$$\mathcal{L}_{\text{recovery}} = \mathcal{L}_{\text{diffusion.noise.mse}} + \lambda \|\epsilon_{\text{student}} - \epsilon_{\text{teacher}}\|_2^2. \quad (4)$$

The first term trains the student on the PianoMime low-level diffusion task. The second term makes the student noise prediction close to the frozen official low-level teacher. This term guides the RP1M-derived student toward the official PianoMime low-level behavior distribution during recovery.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

We organize the experiments into baseline reproduction, specialist improvement, generalist improvement, and RP1M diagnostics. The single-song baseline clips are Happy_8,

Table 2: Single-song PianoMime baseline reproduction.

Clip	F1
Happy_8	0.8724
ImagineDragons_6	0.9158
LetMeDownSlowly_6	0.9289

Table 3: Specialist improvement on Happy_8.

Method	F1	Δ F1
Baseline	0.8725	–
Onset + light smoothness + residual calibration	0.9187	+0.0463

ImagineDragons_6, and LetMeDownSlowly_6. The specialist improvement uses Happy_8 as the fixed comparison clip. The multi-song generalist benchmark uses five unseen clips from dataset/notes_test: Alone_1, EyesClosed_1, Hope_1, SomewhereOnlyWeKnow_1, and NoTimeToDie_1. All generalist methods are compared under the same five-clip protocol. We report precision, recall, F1, learning curves, and performance videos.

4.2 BASELINE REPRODUCTION

Table 2 reports the reproduced single-song specialist baselines. These results verify that the original PianoMime single-song setting runs successfully on the selected clips. The generalist baseline is reported together with the generalist modifications in Table 4.

4.3 SPECIALIST COMBINED METHOD

Table 3 reports the specialist result on Happy_8. The combined method improves F1 from 0.8725 to 0.9187, with a gain of 0.0463. The improved rollout reaches precision 1.0000 and recall 0.8895 in our summary. The baseline precision and recall are not included in the current single-song summary table, so the main table reports F1 and Δ F1.

4.4 GENERALIST POLICY MODIFICATIONS

Table 4 reports the main generalist comparison on the five unseen clips. The motion-weighted low-level diffusion fine-tune achieves the best average F1 among the merged generalist results, increasing mean F1 from 0.6150 to 0.6258. The existing positive generalist modifications are retained for comparison: the smooth denoised diffusion objective increases mean F1 to 0.6256, goal side-channel representation increases mean F1 to 0.6235, and reward-based rollout selection increases mean F1 to 0.6196.

Figure 2 visualizes the motion-weighted low-level diffusion sweep from the available five-clip evaluation CSVs. The sweep is non-monotonic, with the best mean F1 at $\alpha = 3$. The per-clip panel shows that the mean improvement still hides substantial clip-level variation.

4.5 COMPACT DIAGNOSTIC OUTCOMES

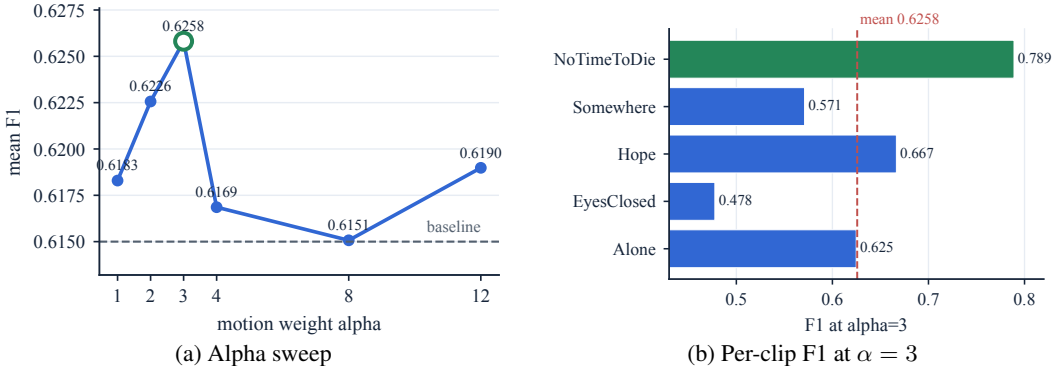
Table 5 summarizes negative and inconclusive findings that are useful for interpreting the positive results. We keep this table compact in the main text and defer exact values, CSV files, and artifact paths to Appendix A.

4.6 ADDITIONAL POSITIVE EVIDENCE

In the broader RPIM evaluation, aligned low-level replacement improves Forester_1 from 0.6731 to 0.6986. Teacher-regularized recovery reaches about 0.6897 on Forester_1. These positive single-clip diagnostics indicate that external robot piano data can help after interface alignment. At the same time, the current five-clip RPIM averages remain below the official baseline,

Table 4: Main generalist results on five selected unseen clips.

Method	Precision	Recall	F1	$\Delta F1$
Baseline	0.7000	0.6156	0.6150	—
Motion-weighted Low-level Diffusion Fine-tune	—	—	0.6258	+0.0108
Smooth Diffusion Objective	0.7010	0.6248	0.6256	+0.0105
Goal Representation	0.7064	0.6254	0.6235	+0.0085
Reward-based Rollout Selection	0.7117	0.6132	0.6196	+0.0046

Figure 2: Motion-weighted low-level diffusion fine-tuning on the five unseen clips. The best sweep point is $\alpha = 3$, with mean F1 0.6258 against the merged-paper baseline 0.6150.

so we treat them as diagnostics rather than main positive results and report the detailed numbers in Appendix A.

5 DATA SCALE-UP WITH RP1M

The RP1M experiments are discussed separately because they study a different question from the main reward, smoothness, conditioning, and inference modifications. The main question here is whether larger robot piano-playing demonstrations can be directly used to improve a PianoMime-style generalist policy.

5.1 PIANO MIME AND RP1M DATA SIGNALS

PianoMime starts from paired internet piano videos and MIDI files. The video provides human fingertip motion, and the MIDI file provides piano-roll goals. PianoMime then uses inverse kinematics to obtain nominal Shadow Hand trajectories and trains song-specific residual RL experts. The expert rollouts provide robot states, goals, fingertip trajectories, and actions for the hierarchical generalist policy. This data path is naturally matched to the PianoMime high-level and low-level policy interfaces.

RP1M starts from music pieces and target key sequences. It uses optimal-transport-based fingering to assign fingers to target keys, trains song-specific RL specialists, and collects their robot state-action rollouts. RP1M therefore contains robot action signals and covers many pieces and expert trajectories. Its scale can provide broader state-action coverage than a small collection of PianoMime expert rollouts.

5.2 WHY DIRECT LOW-LEVEL REPLACEMENT IS UNSTABLE

RP1M Low-level Replacement directly uses RP1M-derived data or models in the PianoMime generalist pipeline. This creates an interface problem. In the PianoMime pipeline, the high-level policy outputs PianoMime-style future fingertip trajectories, and the low-level policy predicts PianoMime-style 46-dimensional hand actions from fingertip targets and proprioception. RP1M-derived tra-

Table 5: Compact diagnostic outcomes. Detailed numbers and artifacts are in Appendix A.

Diagnostic	Outcome	Takeaway
Naive action smoothing	Worse	Contact timing is fragile
More DDPM steps	Worse	Sampler mismatch
Per-hand residual gate	No clear gain	Granularity too coarse
Song feature conditioning	No gain	Future-note observations already carry local cues
No-teacher RP1M recovery	Worse	Teacher keeps interface alignment

jectories may use different fingertip ordering, finger-key coding, action coordinates, joint ordering, normalization, control semantics, and expert motion style.

This mismatch can appear even when tensor shapes are compatible. A dimension can have the right numerical shape while representing a different finger, coordinate, or control variable. Direct replacement therefore risks training a model that has learned useful RP1M motion patterns but does not execute them correctly in the PianoMime low-level interface. This explains why the RP1M Low-level Replacement result decreases on the five-clip benchmark.

5.3 HOW TEACHER-REGULARIZED RECOVERY HELPS

RP1M Recovery + Teacher Regularization adds a recovery stage after the RP1M-derived low-level model. The recovery stage uses original PianoMime low-level data, which brings the student back to the PianoMime condition-to-action interface. The teacher term uses the frozen official low-level model as a behavior reference. The recovery loss is

$$\mathcal{L}_{\text{recovery}} = \mathcal{L}_{\text{diffusion_noise_mse}} + \lambda \|\epsilon_{\text{student}} - \epsilon_{\text{teacher}}\|_2^2. \quad (5)$$

The diffusion noise MSE trains the student on the PianoMime low-level denoising task. The teacher MSE makes the student noise prediction close to the official low-level teacher on the same input. This regularization does not create a new set of teacher-labeled actions. It makes the RP1M-derived student learn the teacher’s denoising behavior under the PianoMime interface.

The five-clip average remains below the baseline, while the broader diagnostic result on `Forester_1` shows positive evidence. Aligned replacement improves `Forester_1` from 0.6731 to 0.6986. Teacher-regularized recovery reaches about 0.6897 on `Forester_1`. Removing the teacher lowers F1 to 0.6518. These results indicate that RP1M carries useful external motion information after alignment and recovery, and that the teacher term helps keep the student near the official PianoMime behavior distribution.

5.4 IMPLICATION FOR DATA SCALING

The RP1M study shows that scaling data is difficult in this task. Larger robot demonstration datasets provide more pieces, more state-action rollouts, and more finger-key patterns. They also introduce schema mismatch and distribution mismatch when their representation does not match the target low-level policy interface. Data scale therefore needs interface alignment. For PianoMime-style policies, this alignment includes fingertip format conversion, action or hand-state mapping, recovery on PianoMime low-level data, and teacher-based behavior regularization.

6 DISCUSSION AND CONCLUSION

The specialist experiments show that note onset frames deserve explicit attention. The combined method improves `Happy_8` from 0.8725 to 0.9187 F1. This result supports the design of onset-aware reward shaping. It also shows that residual strength matters in the specialist pipeline. The residual factor controls the amount of learned correction added to the nominal IK action, and calibration helps the final execution.

The generalist experiments show that motion stability can improve policy quality when it preserves contact timing. Motion-weighted low-level diffusion fine-tuning gives the strongest mean F1 among the merged main generalist results, increasing the five-clip average from 0.6150 to 0.6258. Smooth

denoised diffusion training remains a positive comparison result at 0.6256 mean F1. The diagnostic results clarify why the positive variants work. Smoothness is helpful only when imposed in a timing-aware training objective. Post-hoc low-pass filtering smooths motion visually but shifts or suppresses key contacts, reducing recall. Increasing DDPM denoising steps also fails to help when the inference schedule no longer matches the trained sampler.

The residual-gating and conditioning diagnostics suggest that conditioning must have the right granularity. Per-hand residual gates are too coarse and remain within PPO noise, while per-finger onset-aware gates improve the matched-budget specialist comparison. Coarse song-level statistics also do not improve zero-shot generalization. This suggests that the useful conditioning signal is local, per-finger, and onset-dependent rather than global per-song metadata; the existing future-note observations already contain much of the needed local structure.

The goal-conditioning and inference results show two additional directions for generalist policies. Goal side-channel representation improves the low-level policy by giving it more explicit future-note information. Reward-based rollout selection improves inference by using the stochasticity of diffusion sampling. One component changes the training-time conditioning signal. The other component changes the inference-time selection rule. Both improve the five-clip average under the shared evaluation protocol.

The RP1M analysis shows that large-scale robot piano data requires representation alignment before it can reliably improve a PianoMime-style low-level policy. RP1M provides broader robot demonstration coverage, while direct low-level replacement and no-teacher recovery can drift away from the PianoMime action/state interface. The recovery experiment shows a partial solution: train on PianoMime low-level data and regularize the RP1M-derived student toward the frozen official low-level teacher. The broader *Forester_1* diagnostic results support this direction, while the five-clip benchmark shows that the current alignment is still incomplete. Teacher regularization and learned representation mapping are therefore necessary for stable transfer.

Overall, we reproduce the PianoMime specialist and generalist baselines and evaluate targeted modifications for dexterous piano playing. The experiments verify several practical insights. Note onset frames are important for specialist F1. Motion stability helps generalist execution when contact timing is preserved. Residual calibration affects the balance between IK priors and learned corrections. Explicit future-note conditioning improves low-level action prediction. Stochastic diffusion rollouts can be selected at inference. Large-scale external demonstrations are difficult to scale up directly because the action representation, fingertip format, and behavior distribution must match the target policy interface. Future work should focus on stronger goal representations, contact-aware diffusion objectives, and explicit data-alignment modules for large-scale robot piano demonstrations.

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A DIAGNOSTIC RESULTS WITHOUT FIVE-CLIP AVERAGE GAINS

This appendix reports diagnostic experiments that do not improve the main five-clip average F1, while still helping interpret the method design. We keep these results separate from the main text so that the body of the paper emphasizes methods with positive average improvements, except for onset-related specialist analysis. Each subsection lists the setting, result, artifact path, and short interpretation.

A.1 NAIVE SMOOTHING AND DDPM-STEP DIAGNOSTICS

Setting. We tested direct post-processing smoothness proxies, including low-level action smoothing and high-level demonstration smoothing, as well as increasing low-level DDPM inference from 50 to 75 steps.

Table 6: Timing-sensitive generalist diagnostics.

Variant	F1	Interpretation
Low-level DDPM, 50 steps	0.6914	Reference schedule
Low-level DDPM, 75 steps	0.6629	More steps degraded timing
Low-level action smoothing, $\alpha = 0.2$	0.5939	Delayed or suppressed contacts
Low-level action smoothing, $\alpha = 0.4$	0.5230	Stronger contact degradation
High-level demo smoothing, window 3	0.4783	Fingertip timing blurred

Artifacts. The smoothing and DDPM-step summaries are in `results/improvements/README.md`; related files are under `results/improvements/forester1_smoothing/`.

Interpretation. These results show that visually smoother trajectories are not necessarily better for note-level F1. Piano execution is timing-sensitive, and post-hoc smoothing can shift or weaken key contacts.

A.2 RESIDUAL-GATE GRANULARITY DIAGNOSTICS

Setting. We compared a per-hand residual gate with a per-finger onset-aware residual gate. The per-hand gate emits two gates, one per hand. The per-finger gate emits ten gates, one for each finger, so the policy can modulate residual strength at the finger responsible for an onset.

Table 7: Residual-gate granularity diagnostics.

Variant	Seed 0	Seed 1	Mean
Vanilla matched-budget PPO	0.8124	0.7917	0.8020
Per-finger onset-aware gate	0.8223	0.8105	0.8164
Per-hand residual gate	–	–	No decisive gain

Artifacts. Code is in `single_task/wrappers/residual.py`; curves are in `results/kiro_single_song/glpf_{van,pf}_s{0,1}/eval_metrics.csv`; the summary figure is `results/kiro_figs/goall_perfinger_vs_vanilla.png`.

Interpretation. Per-hand control is too coarse for note onsets. The useful residual-control signal is local and finger-specific.

A.3 SONG-FEATURE CONDITIONING DIAGNOSTICS

Setting. We trained a multi-song per-finger-gate policy with and without a four-dimensional song feature channel: note density, polyphony, onset rate, and active fraction. The policy was evaluated zero-shot on held-out songs.

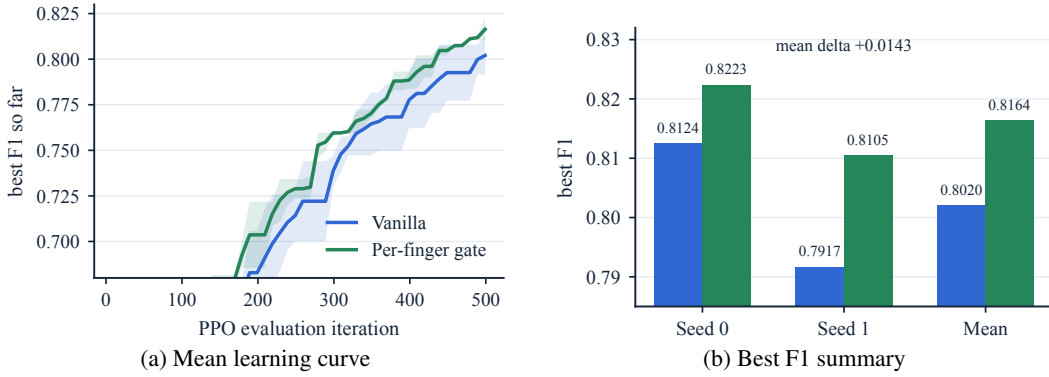


Figure 3: Matched-budget residual-gate comparison from the available specialist evaluation CSVs. The per-finger gate improves the two-seed mean best F1 over the vanilla residual policy.

Table 8: Song-level conditioning diagnostics on held-out songs.

Variant	Mean held-out F1
With song features	0.5466
Without song features	0.5531

Artifacts. Plotting data are the two multi-song evaluation CSV files under `paper_assets/data/`. The corresponding implementation is in the multi-song trainer and song-feature wrapper.

Interpretation. Coarse whole-song metadata did not improve zero-shot generalization. The policy already observes future notes, and the useful conditioning appears to be local, temporal, and per-finger rather than global per-song.

A.4 REWARD-SELECTED ROLLOUT DIAGNOSTIC

Setting. We sampled three low-level diffusion rollouts per clip and selected the rollout with the highest episode reward. This is a test-time selection diagnostic rather than a training improvement.

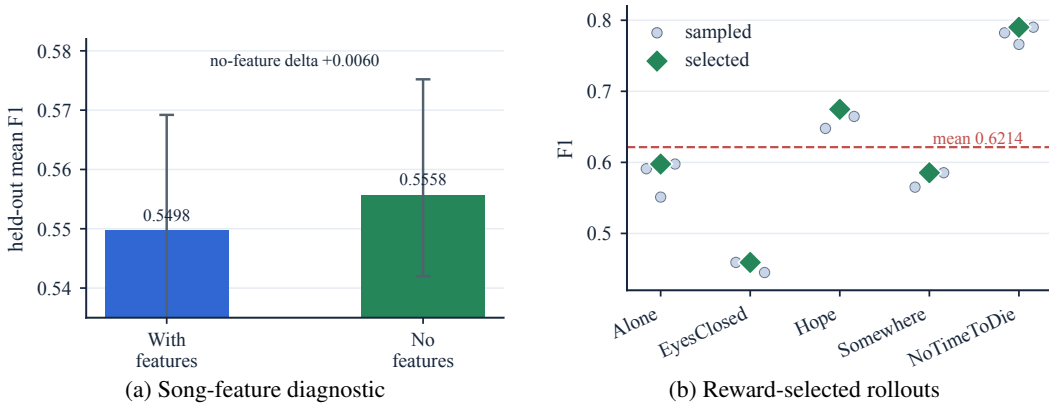


Figure 4: Additional compact diagnostics from existing CSVs. Left: coarse song features do not improve held-out F1. Right: best-of- N selection reports the reward-selected rollout rather than the raw average over all sampled rollouts.

Interpretation. Reward selection recovers a higher five-clip mean F1 than the raw average over all sampled rollouts, but it should be interpreted as a test-time selection result rather than evidence of a better trained policy.

A.5 RP1M FIVE-CLIP DIAGNOSTICS

Table 9 reports RP1M-related results under the same five-clip benchmark. RP1M Recovery + Teacher Regularization reaches 0.6094 mean F1, and RP1M Low-level Replacement reaches 0.5956 mean F1. These two five-clip averages are lower than the baseline. They are kept as diagnostics for the data-scale and representation-alignment analysis in Section 5.

Table 9: RP1M diagnostic results on five selected unseen clips.

Method	Precision	Recall	F1	Δ F1
RP1M Recovery + Teacher Reg.	0.7051	0.6027	0.6094	-0.0056
RP1M Low-level Replacement	0.6811	0.5934	0.5956	-0.0194

Artifacts. Five-clip RP1M diagnostics are summarized in results/table9_eval/ac_pandora_report.md; metrics include results/table9_eval/official_ddpm_seed0/metrics.csv, results/table9_eval/ac_recover100_teacher0005_seed0/metrics.csv, and results/table9_eval/ac_recover50_teacher0005_seed0/metrics.csv.

Interpretation. RP1M contains useful motion information, but the five-clip average shows that direct transfer remains unstable without stronger interface alignment.

A.6 NO-TEACHER RP1M RECOVERY

Setting. We compared teacher-regularized recovery against recovery without the frozen official low-level teacher on Forester_1.

Table 10: Teacher regularization diagnostic on Forester_1.

Variant	Precision	Recall	F1
A+C recover50 + teacher 0.005	0.8032	0.7310	0.6897
A+C recover100 + teacher 0.005	0.7970	0.7336	0.6974
A+C recover100, no teacher	0.7807	0.6931	0.6518

Artifacts. Results are summarized in results/current_repo_improvements_and_assets.zh.md and results/all_experiments_summary.md; the corresponding video artifact includes results/improvements/twostage_linearq_recover100_teacher0.005_seed3.mp4.

Interpretation. Removing the teacher allows the recovered student to drift away from the PianoMime low-level action/state interface. Teacher regularization stabilizes transfer by keeping the RP1M-derived student near the official behavior distribution.

A.7 LONG REWARD-SHAPED SPECIALIST FINE-TUNE

Setting. We continued training a strong full-budget Happy_8 specialist with light smoothness and onset reward shaping.

Table 11: Long reward-shaped specialist fine-tune diagnostic on Happy_8.

Variant	Precision	Recall	F1
Full-budget residual PPO	1.0000	0.8895	0.9187
Long reward-shaped fine-tune	0.9406	0.7759	0.8057

Artifacts. The detailed report is results/happy8_single_song_homework_report.md; related summaries are in results/single_song_course_report.md.

Interpretation. Reward shaping can improve early-stage learning or sample efficiency, but long fine-tuning from an already strong specialist can distort the optimized behavior and reduce recall.

A.8 ADDITIONAL NEGATIVE DIAGNOSTICS

We also keep the full alpha and seed sweeps, high-level RP1M fine-tuning checkpoints, smoothing-window tables, released-rollout residual sweeps, and reward-selection candidate CSVs as artifacts rather than expanding them in the main text. These files are useful for reproducibility, but the main paper focuses on the few diagnostics that explain the positive results.